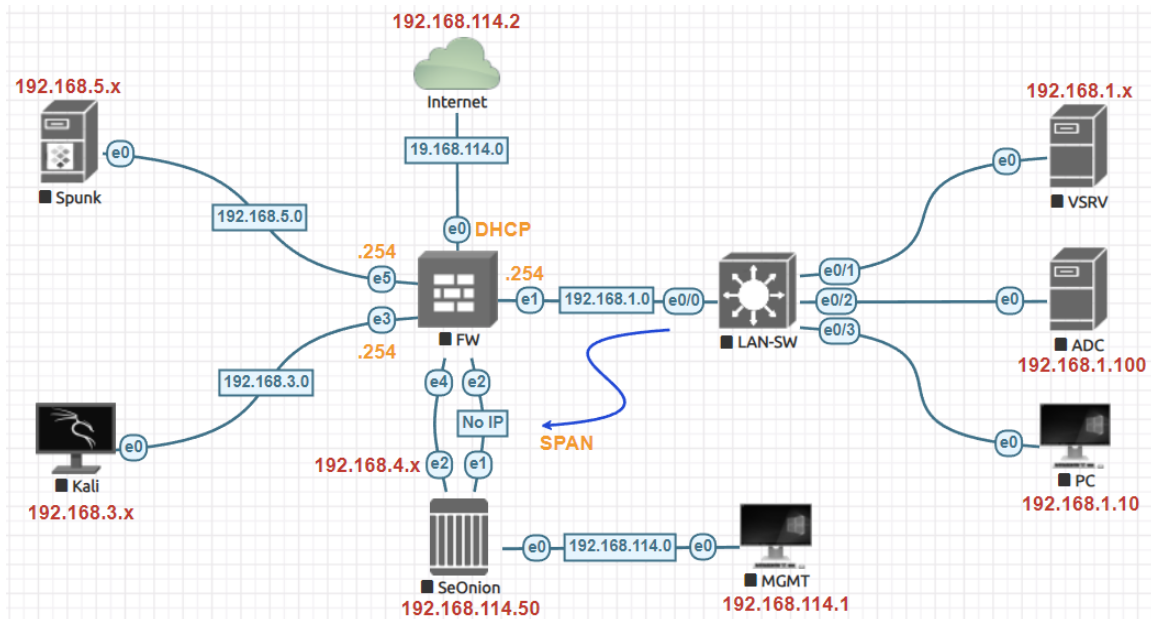


John Brown Alhassan — Cybersecurity Portfolio

SOC Analyst | Blue Team | Threat Detection | Incident Response

Nmap Network Reconnaissance and Detection Lab Report

This report documents a complete end-to-end security analysis performed in a controlled enterprise lab environment designed and engineered by John Brown. The lab reproduces a realistic corporate network consisting of multiple segmented VLANs, a pfSense firewall, a Windows Active Directory domain, a Kali attack host, a Security Onion detection stack, a Splunk SIEM instance, and a centralized management workstation.



The network layout—shown in the attached topology diagram—illustrates the flow of traffic between subnets:

- **192.168.1.x** (Server VLAN – AD/DC, File/Service Hosts)

- **192.168.3.x** (Kali Attacker VLAN)
- **192.168.4.x** (Security Onion Sensor VLAN)
- **192.168.5.x** (Splunk SIEM VLAN)
- **192.168.114.x** (Management + Internet-facing subnet)

The objective of this project is to simulate a realistic adversarial reconnaissance attack using **Nmap** from the Kali machine (192.168.3.x) against the Windows Server Domain Controller (192.168.1.100), while simultaneously monitoring and analyzing the resulting traffic across:

- **Wireshark** (packet-level analysis on Kali)
- **Security Onion** (Zeek session logs + Suricata alerts)
- **pfSense Firewall** (state table + rule hit indicators)

This investigation demonstrates:

1. **How real-world reconnaissance appears on the wire** using SYN scans, service enumeration, and aggressive OS fingerprinting.
2. **How a modern SOC toolset detects and correlates scanning behavior**, shown through Suricata signatures, Zeek connection logs, and packet captures.

The results show clear detection of Nmap's scanning patterns, including:

- Suricata signature hits such as *"ET SCAN Nmap Scripting Engine Detected"*
- Zeek conn.log entries showing rapid sequential port probes
- Wireshark evidence of SYN flood-like behavior and ICMP probing
- Firewall path visibility confirming permitted east-west traffic

2. Evidence and Analysis

2.1 Kali Network Configuration and Reachability

The first step was to confirm the Kali attack machine's IP configuration and its ability to reach the domain controller.

- Figure 1 shows 'ifconfig' output confirming eth0 is configured with 192.168.3.1/24.
- Figure 2 shows a successful ICMP ping from 192.168.3.1 to 192.168.1.100, proving routing and basic connectivity before launching Nmap.

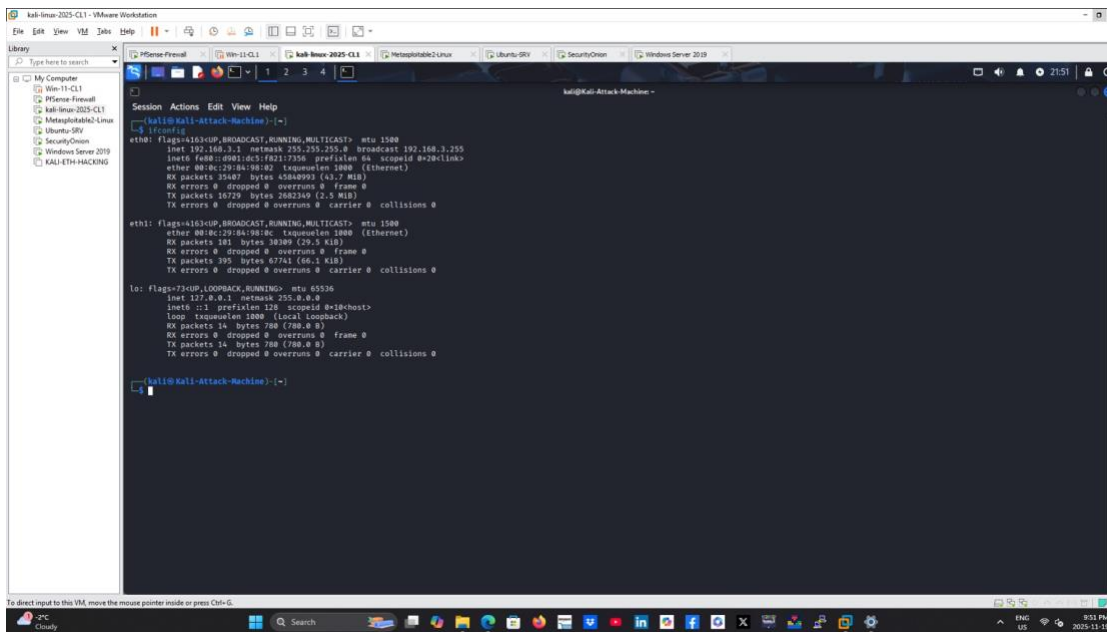


Figure 1: Kali ifconfig output showing the attack host on 192.168.3.1 (eth0) and confirming basic connectivity.

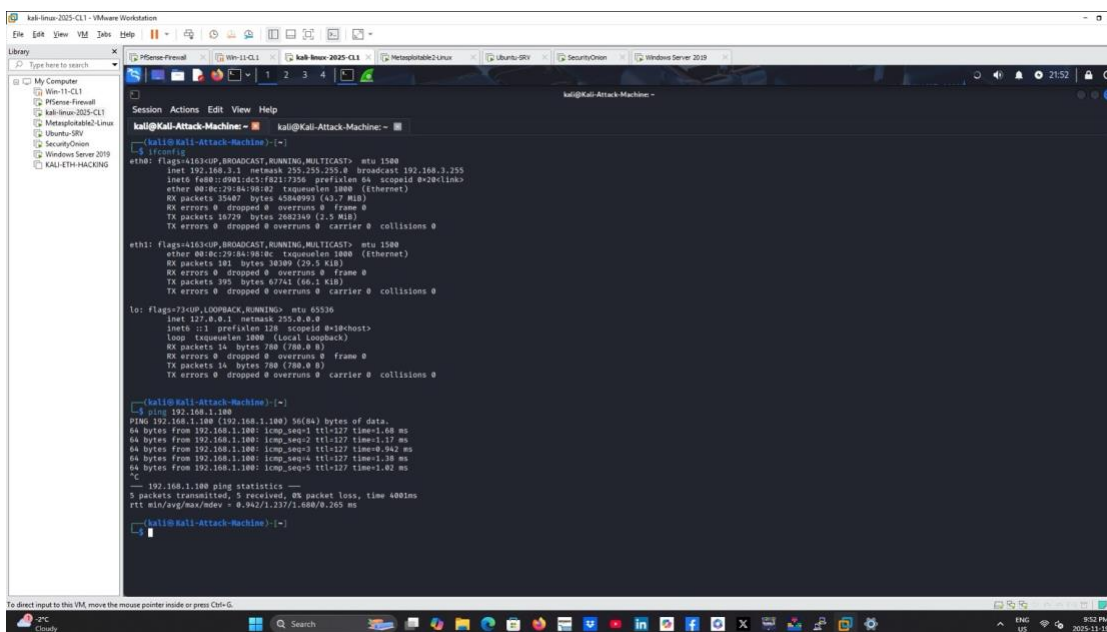


Figure 2: Kali ping test from 192.168.3.1 to the domain controller 192.168.1.100 verifying ICMP reachability before scanning.

2.2 Nmap Reconnaissance Results

With connectivity verified, Nmap was used to enumerate open ports and services on the domain controller.

- Figure 3 captures the initial Nmap scan summary, confirming that the host is up and identifying several open TCP ports.
- Figure 4 shows the full '-A' aggressive scan, which performs version detection, OS fingerprinting, script scanning, and traceroute. Key Active Directory services such as Kerberos, LDAP, SMB, and RDP are clearly identified.
- Figure 5 continues the Nmap output, including smb2-security-mode script results and OS guesses (Windows 10 / Windows Server 2019).

```

kali@kali-2025-CL1: ~
└─$ nmap -sS 192.168.1.100
Nmap scan report for 192.168.1.100
Host is up (0.0026s latency).
Not shown: 996 open/filtered ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  Microsoft HTTPAPI httpd 2.0 (SSDP/Udp)
139/tcp   open  Microsoft Windows SMB
389/tcp   open  ldap
443/tcp   open  https
464/tcp   open  Microsoft Windows Active Directory LDAP (Domain: test.local, Site: Default-First-Site-Name)
593/tcp   open  hpsa
636/tcp   open  https
3268/tcp  open  ldap
3389/tcp  open  rdp
Nmap done: 1 IP address (1 host up) scanned in 0.01 seconds

```

Figure 3: Initial Nmap scan results against 192.168.1.100 showing open ports and OS detection summary.

```

kali@kali-2025-CL1: ~
└─$ nmap -A 192.168.1.100
Nmap scan report for 192.168.1.100
Host is up (0.0026s latency).
Not shown: 996 open/filtered ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Microsoft HTTPAPI httpd 2.0 (SSDP/Udp)
135/tcp   open  Microsoft Windows RPC
139/tcp   open  Microsoft Windows SMB
389/tcp   open  ldap        Microsoft Windows Active Directory LDAP (Domain: test.local, Site: Default-First-Site-Name)
443/tcp   open  https       Microsoft Windows Active Directory LDAP (Domain: test.local, Site: Default-First-Site-Name)
464/tcp   open  hpsa
593/tcp   open  hpsa
636/tcp   open  https       Microsoft Windows Active Directory LDAP (Domain: test.local, Site: Default-First-Site-Name)
3268/tcp  open  ldap        Microsoft Windows Active Directory LDAP (Domain: test.local, Site: Default-First-Site-Name)
3389/tcp  open  rdp         Microsoft Windows Remote Desktop Services
OS: cpe:/o:microsoft:windows_server:2019
OS CPE: cpe:/o:microsoft:windows_server:2019
OS fingerprint: Microsoft Windows 2019 (97%), Microsoft Windows 10 1803 (91%), Microsoft Windows 10 1809 (91%), Microsoft Windows Server 2019 (91%)
Aggressive OS guesses: Windows Server 2019 (97%), Microsoft Windows 10 1803 (91%), Microsoft Windows 10 1809 (91%), Microsoft Windows Server 2019 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops
Service Info: Host: SVR; OS: Windows; CPE: cpe:/o:microsoft:windows

```

Figure 4: Full Nmap -A scan against 192.168.1.100 enumerating key Windows AD services such as Kerberos (88), LDAP (389/636), SMB (445) and RDP (3389).

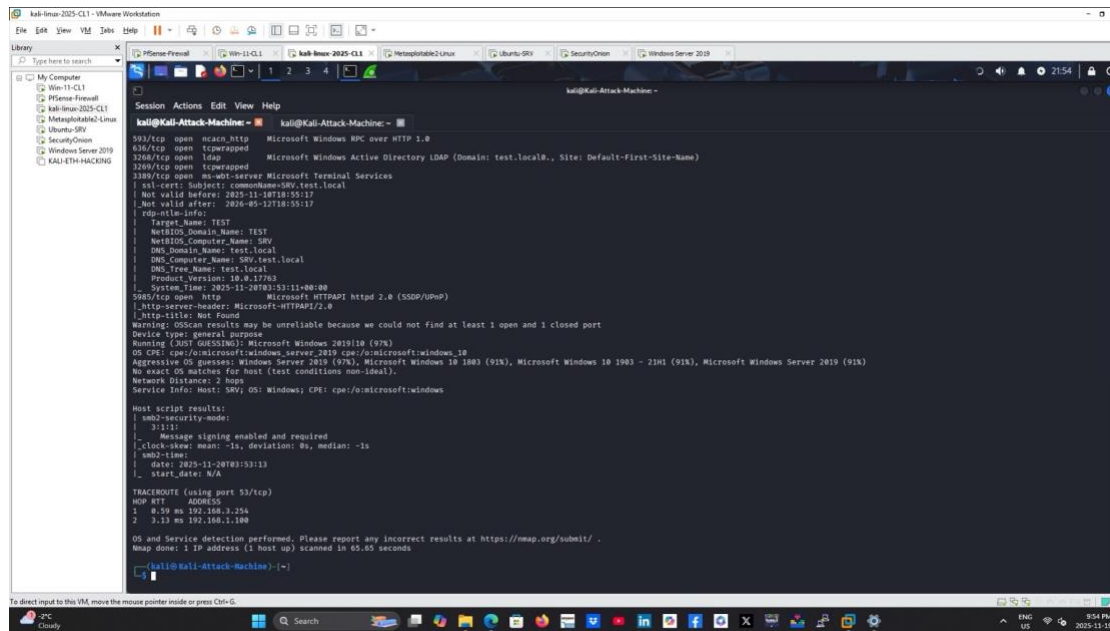


Figure 5: Continuation of Nmap service and OS detection output, including smb2-security-mode script results and network distance.

2.3 Wireshark Traffic Analysis on Kali

During the Nmap scan, Wireshark was running on Kali to capture packet-level details.

- Figure 6 applies the 'tcp.flags.syn==1 && tcp.flags.ack==0' filter, showing the outbound SYN probes from Kali to many destination ports on 192.168.1.100. This is the classic signature of a TCP connect or SYN scan.
- Figure 7 focuses on ICMP traffic, including echo replies and 'Destination unreachable (Port unreachable)' messages that Nmap uses to infer host and service reachability.
- Figure 8 uses the 'tcp.flags.syn==1 && tcp.flags.ack==1' filter to highlight SYN/ACK responses from the domain controller, which Nmap interprets as open ports.
- Figure 9 adds a UDP-focused filter 'udp && ip.src == 192.168.3.1' to show background DNS, NTP, and NetBIOS traffic generated by the domain controller during the scan window.

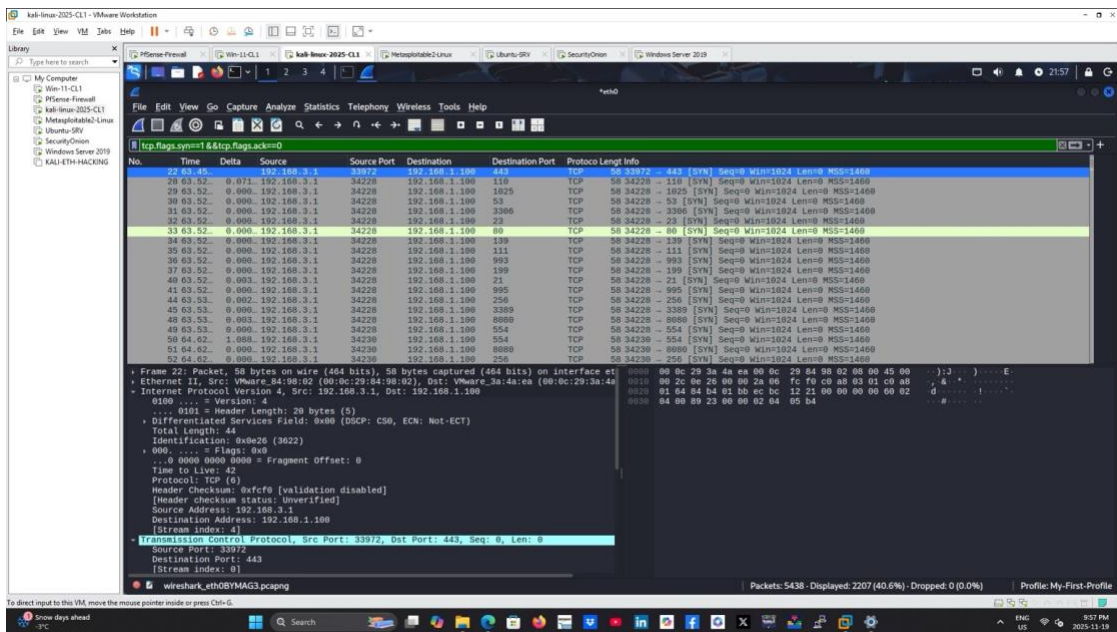


Figure 6: Wireshark capture on Kali filtered with 'tcp.flags.syn==1 && tcp.flags.ack==0' highlighting outbound SYN packets from 192.168.3.1 to multiple ports on 192.168.1.100 during the Nmap scan.

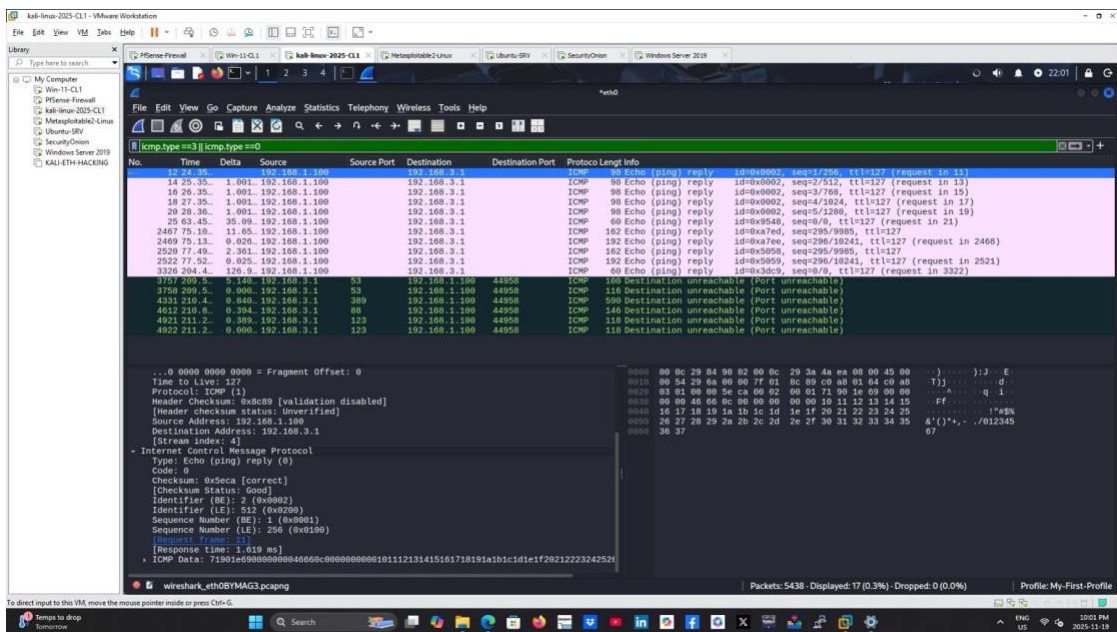


Figure 7: Wireshark ICMP view filtered to show echo replies and ICMP Destination Unreachable messages, illustrating how the target responds to Nmap host discovery and closed ports.

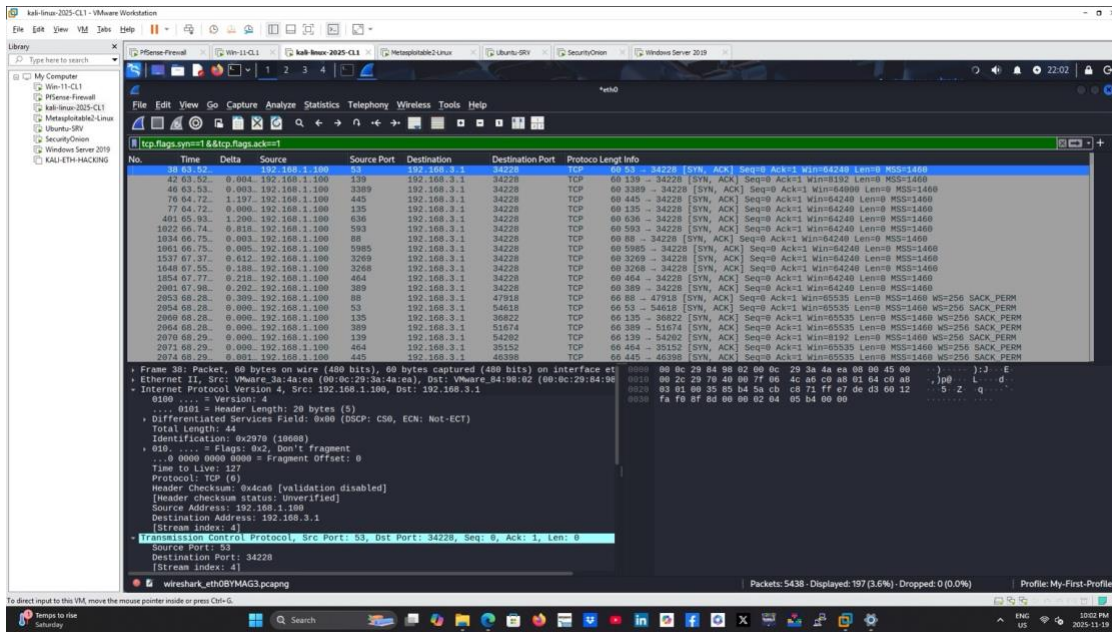


Figure 8: Wireshark TCP response traffic filtered with 'tcp.flags.syn==1 && tcp.flags.ack==1' to highlight SYN/ACK packets from 192.168.1.100 back to 192.168.3.1 for open ports discovered by Nmap.

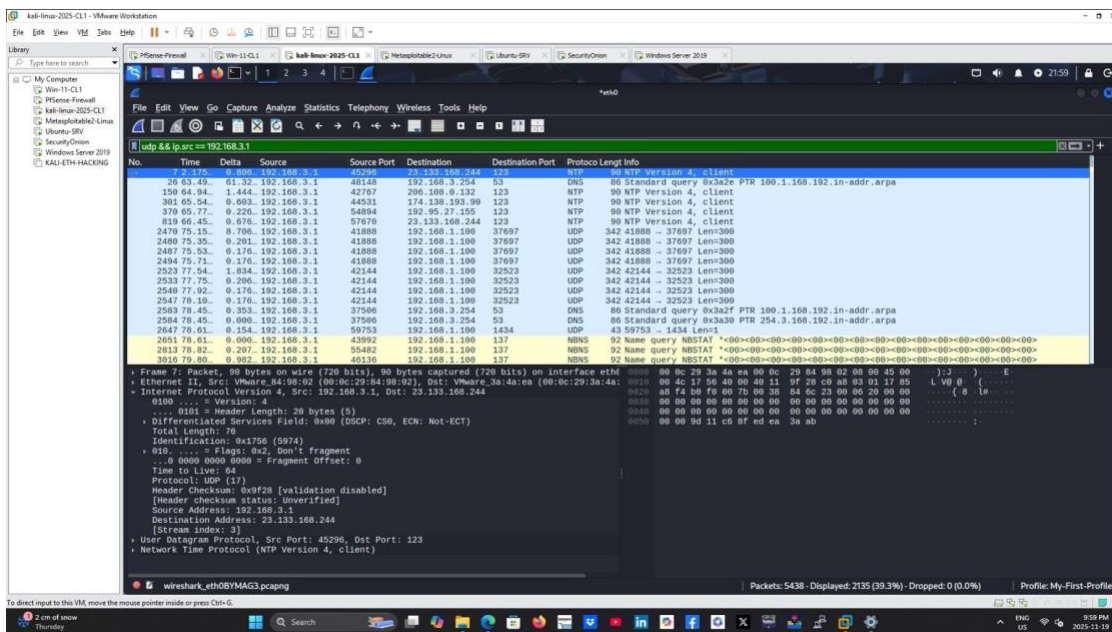


Figure 9: Wireshark UDP filter 'udp && ip.src == 192.168.3.1' focusing on DNS, NTP, and NetBIOS traffic originating from the domain controller as background noise during the scan.

2.4 Security Onion Detection (Suricata & Zeek)

Security Onion monitors the same network using Suricata for intrusion detection and Zeek for protocol analysis.

- Figure 10 shows Suricata alerts triggered by the Nmap activity, including rules such as 'ET SCAN Nmap Scripting Engine User-Agent Detected' and 'ET SCAN Possible Nmap User-

Agent Observed'. These confirm that the sensor can reliably detect Nmap reconnaissance.

- Figure 11 presents the Zeek Hunt interface filtered to show 'zeek.conn' events where the source IP is 192.168.3.1. Each row corresponds to a connection attempt from Kali to SRV, providing a high-fidelity log of the scan at the session layer.

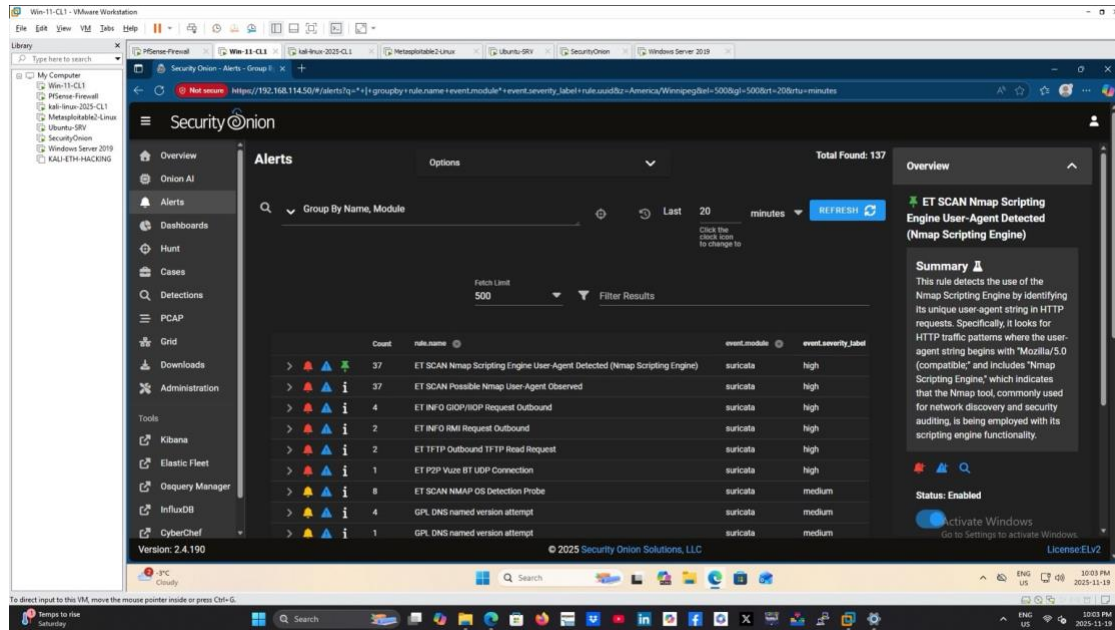


Figure 10: Security Onion Suricata Alerts view showing multiple high-severity signatures triggered by the Nmap activity (e.g., ET SCAN Nmap Scripting Engine User-Agent Detected).

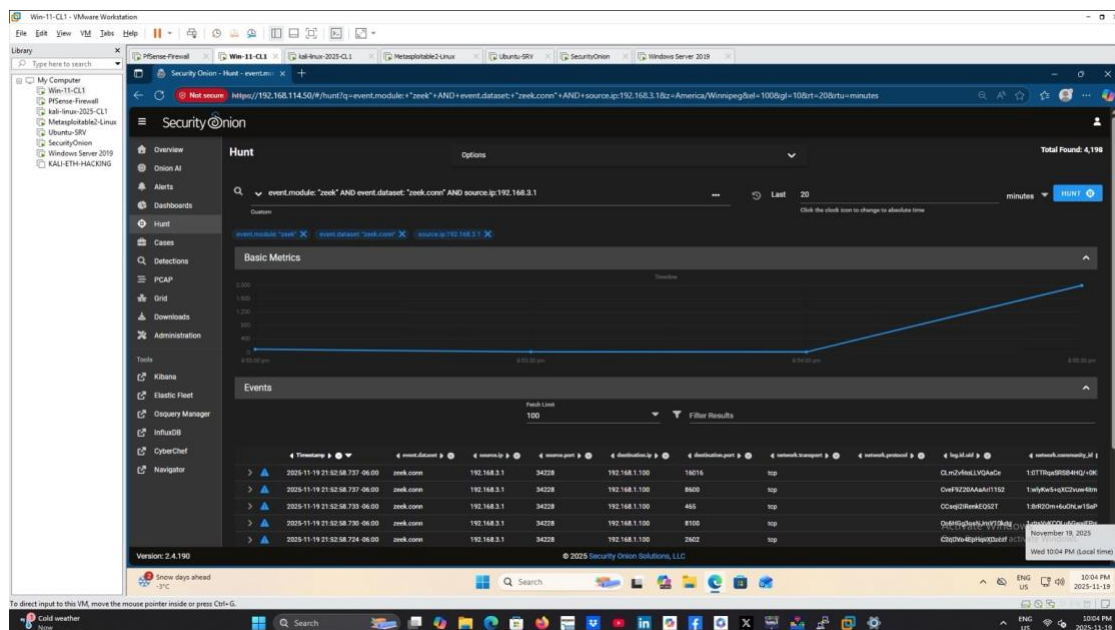


Figure 11: Security Onion Zeek Hunt view filtered for 'event.module:"zeek" AND event.dataset:"zeek.conn" AND source.ip:192.168.3.1', listing Zeek connection logs that map each Nmap probe from the attacker to the domain controller.

3. Conclusion

This lab demonstrates an end-to-end view of Nmap reconnaissance in a Windows domain environment. The Kali attack machine generated scan traffic, which was captured in Wireshark and detected by Security Onion's Suricata IDS and Zeek. Together, these tools provide strong visibility into network probing activity, allowing defenders to identify hostile reconnaissance early in the attack lifecycle.

Thank you

John Brown